

WASP-43b

R-0611

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_180316 · created 2026-07-25T07:44:00+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458193.8096 +/- 0.0012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.178 +/- 0.014
Transit depth (Rp/Rs)^2	3.17 +/- 0.51 [%]
Orbital Inclination (inc)	81.13 +/- 0.84
Airmass coefficient 1 (a1)	1.0205 +/- 0.0097
Airmass coefficient 2 (a2)	-0.015 +/- 0.01
Scatter in the residuals of the lightcurve fit is	0.94 %
Best Comparison Star	#3 - [338, 115]
Optimal Aperture	3.51
Optimal Annulus	7.27
Transit Duration (day)	0.049 +/- 0.0032

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.178 ± 0.014 vs 0.1594 (deviation 1.33σ)
Duration fit vs published	0.049 d vs 0.0512 d (deviation 0.7σ)
Mid-time O–C	-0.2 min
Depth SNR	12.7
Residual scatter	0.94 %

Light curve

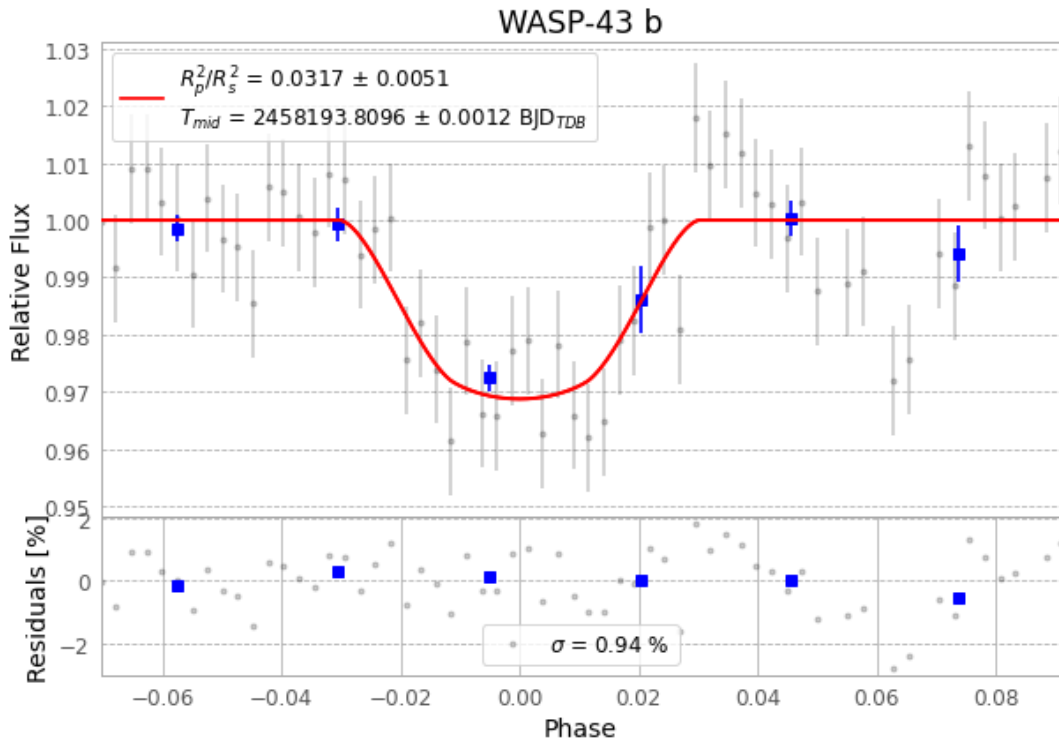


Figure 1 — FinalLightCurve_WASP-43 b_2018-03-15.png (SHA-256 af1877086328c7e4...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [425, 284], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 3defb44fb3d84a8e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

64 FITS frames (42 MB), WASP-43180316055415.FITS → WASP-43180316090312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-180316.log (e7b2912bdbf0...), FinalLightCurve_WASP-43 b_2018-03-15.png (af1877086328...), FinalLightCurve_WASP-43 b_2018-03-15.csv (c7c30068403c...)

Compute resources

Wall time 120.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-25 07:44Z.